

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
S Susmitha	192571068	ITA0510	Computer Vision

21. Perform Sharpening of Image using Laplacian mask implemented with an extension of diagonal neighbors.

1	1	1
1	-8	1
1	1	1

AIM:

Perform Sharpening of Image using Laplacian mask implemented with an extension of diagonal neighbors.

PROGRAM:

```
import cv2
import numpy as np

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png')

if img is None:
    print("Error: Image not found!")
else:
    # Sharpening kernel
    kernel = np.array([
        [1, 1, 1],
        [1, -8, 1],
        [1, 1, 1]
    ])

    # Apply the sharpening filter
    sharpened = cv2.filter2D(img, -1, kernel)

    # Save the output image
    cv2.imwrite(
        r'C:\Users\Susmitha\Downloads\ITA0510 Lab\Sharpened_Image.jpg',
        sharpened
```

)

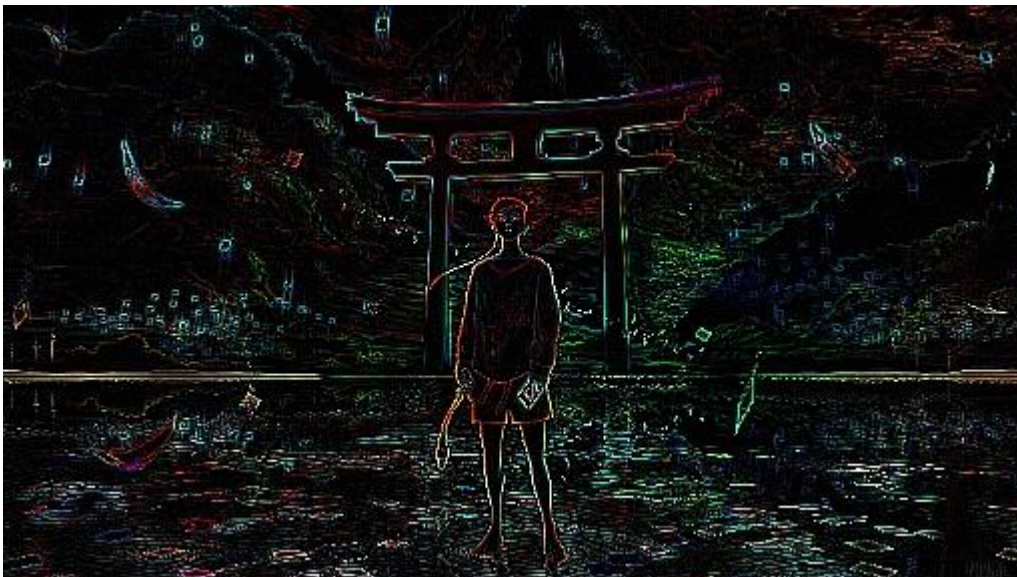
```
# Display the images
cv2.imshow("Original Image", img)
cv2.imshow("Sharpened Image", sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.